

MapLizard

Autonomous Subterranean & Structural Mapping via Cameraless Sensory Swarms

[TEAM NAME] · [REGISTRATION ID] · FALLING SUN 2026 HARDWARE TRACK

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Exploring disaster zones costs lives, land, and hardware



Human endangerment

Surveyors and rescuers enter unstable, zero-visibility, structurally compromised zones.



Toxic leaching

Lost rovers with heavy lithium packs and chemical sensors break down and seep into groundwater.



E-waste

LiDAR and camera-heavy robots are too expensive to risk; one cave-in means non-recoverable electronics.

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MapLizard: a cameraless micro-rover swarm

Zero-optical dependency

Maps unknown terrain without sight, no heavy processing gear.

Eco-conscious build

Low-impact materials and minimal parts, so a lost unit leaches little.

Swarm synergy

Many cheap, lightweight Lizards replace one massive high-waste robot.

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Ultrasonic sensing plus IMU tracking, reset by reflective markers

Continuous dead reckoning recalibrated by passive physical anchors

01

Sense

Ultrasonic & ToF arrays ping perimeter distances across zero-visibility voids.

02

Track

6-DOF IMU dead reckoning computes relative X, Y, Z displacement without GPS.

03

Anchor

Rovers drop non-toxic recycled-foil markers to fix precise physical ground truth.

04

Reset

TCRT5000 IR sensors detect dropped foil to cancel accumulated IMU drift.

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Built from off-the-shelf parts, about 85% cheaper than LiDAR

~85%

Cost reduction vs conventional LiDAR inspection platforms

Flat modular PCB chassis and standard micro-geared drives eliminate costly optics and specialized compute rigs.

Metric	Conventional LiDAR	MapLizard Swarm
Unit cost	$2,500 – $10,000+	< $50 per rover
Expendability	High financial loss	Fully disposable
Toxic leaching	Heavy lithium cells	Minimal, low-hazard
Dust penetration	Optical beam scattering	Immune (acoustic ToF)

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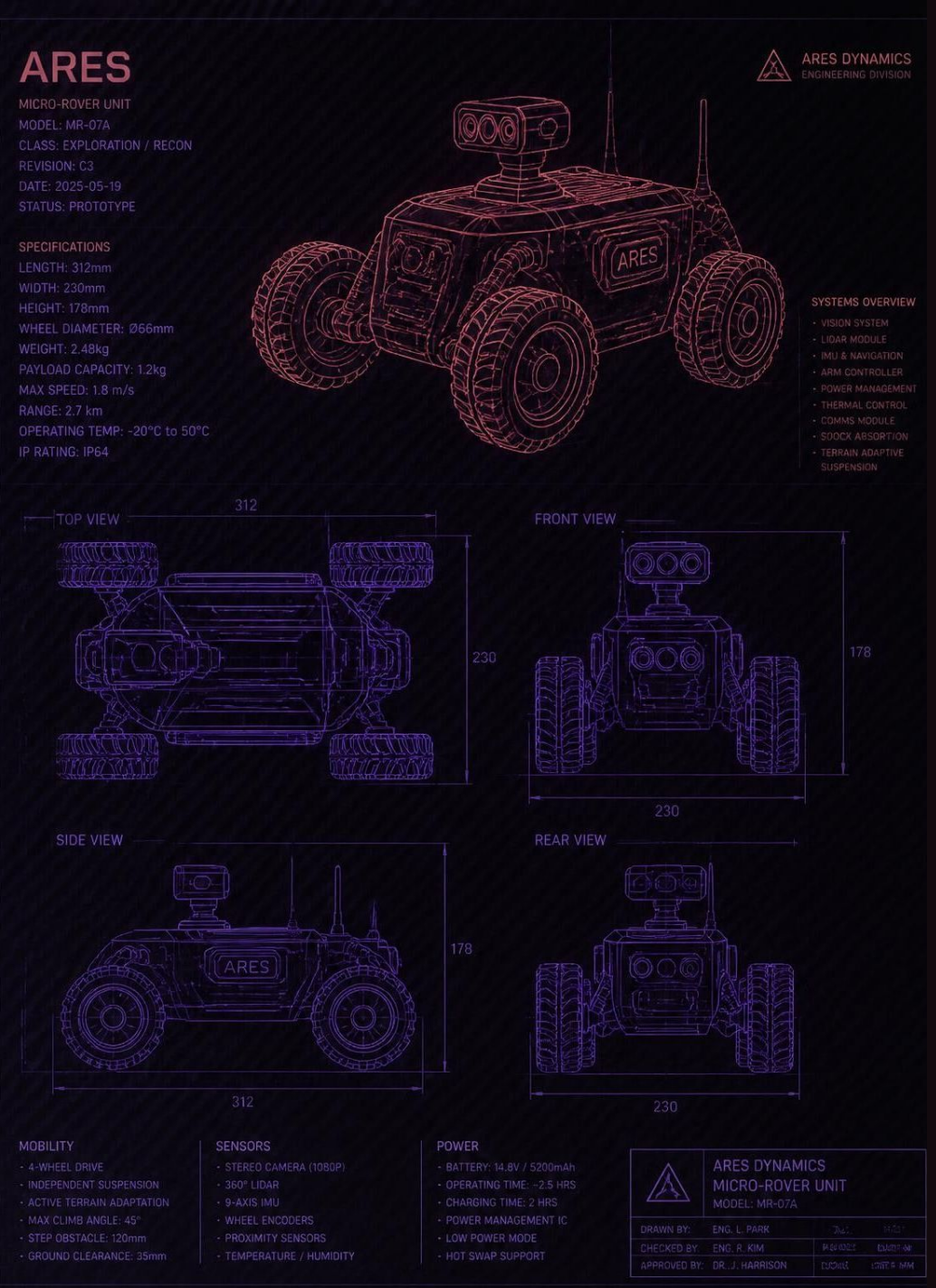
Bill of materials: 15 x 12 x 6 cm, about 350 g

Prototype footprint engineered for tight void and rubble penetration.

Component	Specification/Model	Qty	Purpose
Microcontroller	ESP32 / compact MCU	1	Main logic, sensor reading, Wi-Fi/BLE link
Distance sensors	HC-SR04 / VL53L0X	3	Obstacle detection, wall-following
IMU	MPU6050 (6-DOF)	1	Pitch, roll, yaw, relative displacement
Actuators	N20 micro gear motors	2	High-torque differential drive
Motor driver	L298N Mini / DRV8833	1	Dual H-bridge motor control
Reflective sensor	TCRT5000 IR module	1	Detects reflective foil anchors
Power	3.7V LiPo / LiFePO4	1	Lightweight, reduced leaching risk
Chassis & drops	Eco-plastic + foil	1 set	Chassis frame and passive markers



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Safeguarding human lives, eliminating toxic waste, mapping the unknown

Ready for deployment testing

MapLizard proves that low-cost, expendable sensory swarms can solve extreme subterranean exploration without risking lives or poisoning groundwater.

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